Introduction to Artificial Intelligence

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Abstract

This is the course notes and assignments for the **Introduction to Artificial Intelligence (DSMA121)** course.

Contents

1 AI: A General Introduction **2**
1.1 History and Evolution of AI 2
1.2 AI Application and State of Arts 3
1.3 Societal Impact of AI 3
1.4 Assignment 1 3

Chapter 1

AI: A General Introduction

Lecture 1: First Lecture

Note (Objectives). The class taught:

14 Apr. 19:23

1. Introduction and Definition of AI (connecting to humans and rationality)
2. History and Evolution of AI
3. AI Applications and state of arts
4. Societal impact of AI (jobs, privacy, existential risks)

1.1 History and Evolution of AI

Artificial Intelligence (AI) has a rich history that spans from early theoretical foundations to modern breakthroughs in machine learning and autonomous systems.

Early Foundations-Dark Era (1940s-1950s)

The conceptual groundwork for AI began in the 1940s with Warren McCulloch and Walter Pitts designing the first artificial neurons in 1943. In 1950, Alan Turing published his seminal paper *Computing Machinery and Intelligence*, introducing the Turing Test as a criterion for machine intelligence. The term "Artificial Intelligence" was coined by John McCarthy in 1956 during the Dartmouth Conference, which is widely regarded as the birth of AI as a formal field of study.

Pioneering Developments (1950s-1960s)

In 1951, Marvin Minsky and Dean Edmonds developed the first artificial neural network called SNARC. The 1950s also saw Arthur Samuel create a program that learned to play checkers independently. The 1960s introduced early AI applications such as ELIZA, the first chatbot developed by Joseph Weizenbaum in 1964, and Shakey, the first mobile intelligent robot created by SRI International in 1969. The first industrial robot, Unimate, began working on a General Motors assembly line in 1961.

Challenges and AI Winters (1970s-1980s)

Despite early enthusiasm, AI research faced significant setbacks in the 1970s, including reduced funding following critical reports like James Lighthill's 1973 review. This period, known as

the "AI Winter," saw diminished progress and skepticism. However, the 1980s experienced a revival with the development of expert systems and renewed industrial interest, including early driverless vehicle prototypes by Mercedes-Benz.

Modern Advances (1990s-Present)

The 1990s onward marked rapid advancements in AI capabilities. Notable milestones include IBM's Deep Blue defeating chess champion Garry Kasparov in 1997, the rise of personal assistants and speech recognition technologies, and breakthroughs in deep learning. The 2000s and 2010s saw the emergence of AI-powered applications such as IBM Watson, autonomous vehicles, and sophisticated chatbots like ALICE and GPT-3. More recently, generative AI models have enabled lifelike content creation and AI-driven automation across industries.

Summary Timeline:

- 1943: First artificial neurons by McCulloch and Pitts
- 1950: Turing Test proposed by Alan Turing
- 1956: Term "Artificial Intelligence" coined by John McCarthy
- 1961: Unimate, first industrial robot, deployed
- 1964: ELIZA, first chatbot, developed
- 1969: Shakey, first mobile intelligent robot, created
- 1970s: AI Winter due to reduced funding and skepticism
- 1980s: Revival with expert systems and driverless car prototypes
- 1997: Deep Blue defeats world chess champion
- 2000s-2020s: Rise of personal assistants, deep learning, and generative AI

1.2 AI Application and State of Arts

1.3 Societal Impact of AI

1.4 Assignment 1

Exercise. Choose any possible stream of AI application in Nepal and explore the followings:

1. AI Application domain with Scope, Challenges and Opportunities
2. Possible Risks and risk mitigation strategy
3. Job replacement scenario and how to deal with unemployment
4. Advantages vs limitations lists
5. Government role and social impact

Answer. **AI Application Domain: Agriculture in Nepal**

1. **AI Application Domain with Scope, Challenges, and Opportunities**

Artificial Intelligence in agriculture holds transformative potential for Nepal, a country where a large portion of the population relies on farming. AI applications include precision farming, disease detection, weather and market prediction, and resource optimization. Tools like drones, sensors, and machine learning algorithms can help increase yield and reduce losses.

Scope: Precision agriculture, automated irrigation, pest/disease diagnosis, market analytics, and advisory services using AI chatbots.

Challenges: Lack of digital infrastructure, low AI awareness among farmers, scarcity of localized datasets, high implementation costs, and resistance to adopting new technologies.

Opportunities: Enhanced crop yield, climate-resilient practices, real-time advisory services, empowerment of women farmers, and smart farming systems that make agriculture more efficient and profitable.

2. **Possible Risks and Risk Mitigation Strategy**

   Risks:

   - Data privacy breaches due to inadequate cybersecurity.
   - Algorithmic bias from non-representative datasets.
   - Job displacement in manual farming roles.
   - Dependence on foreign AI tools and platforms.
   - Environmental harm from improper AI-based decisions.

   Mitigation Strategies:

   - Establish strong data governance and privacy regulations.
   - Invest in ethical AI design and transparent algorithms.
   - Provide cybersecurity infrastructure and training.
   - Encourage local AI research to reduce dependency.
   - Mandate environmental impact assessments for AI tools.

3. **Job Replacement Scenario and How to Deal with Unemployment**

   AI tools may reduce the need for manual labor through automation (e.g., AI-powered tractors or robotic harvesting). This could lead to displacement of traditional agricultural workers.

   Strategies to Address This:

   - Retrain workers in digital agriculture, AI system maintenance, and data handling.
   - Promote agri-tech entrepreneurship and startups to create new jobs.
   - Establish vocational training centers focused on AI in agriculture.
   - Provide government grants and microloans for tech-driven agri-businesses.

4. **Advantages vs. Limitations**

Category	Advantages	Limitations
Productivity	Higher yields, efficiency in resource use, automated operations	High initial investment, limited infrastructure
Technology	Real-time insights, disease detection, market predictions	Lack of Nepal-specific data, technical skill gap
Socio-economic Impact	Supports food security, empowers women, rural upliftment	May increase inequality if access is unequal
Sustainability	Reduced chemical use, smart resource allocation	Risk of misuse without proper training or safeguards

5. **Government Role and Social Impact**

   Government Role:

   - Enforce the National AI Policy 2081/2025 with clear goals for agriculture.
   - Invest in digital infrastructure and data centers in rural areas.
   - Offer subsidies and incentives for AI-based farming tools.
   - Facilitate partnerships between research institutions and tech firms.
   - Launch AI education campaigns and pilot programs for farmers.

   Social Impact:

   - Improved food security and income generation for smallholders.
   - Empowerment of marginalized communities, especially women.
   - Potential erosion of traditional farming knowledge if not integrated.
   - Risk of social inequality if digital divide is not addressed.

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Appendix